



Engineering Degree Project

Title of your degree project
- *Optional subtitle*



Author: Your name here

Supervisor: Supervisor name

Lnu Supervisor: Supervisor name

Semester: Spring 202X

Subject: Computer Science

Abstract

The report shall begin with a summary, called abstract. The abstract shall not be longer than a paragraph, and is not divided into more than one piece. It shall contain:

- A short background description to the area of your project
- A description of the problem you investigate
- A motivation why this problem is interesting to investigate
- What you have done to answer the problem
- A short summary of your results

From reading the abstract the reader should clearly understand what the report is all about. The purpose of the abstract is to make the reader interested in continue reading the report, if it covers something that the reader wants to know more about.

Keywords: fill in some keywords for your work here. **Examples:** software architectures, adaptive systems, network intrusion detection, ...

Preface

You can have a preface in the report if you want, but it is not necessary. In this you can write more personal reflections on your degree project. In the preface you can also take the opportunity to thank the people who have been particularly helpful during the report writing, for example if you had any contact with a company that helped with the project, people that guided or helped you during the project, or your family and friends that supported you during the project. The preface shall not be longer than half a page.

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Notice

- Do not change the structure of this template without a discussion (and approval) of your Lnu supervisor.
- Regarding tempus: Stick to *present tense* throughout the report.
- This is a technical report intended for readers with a technical background. Hence, be precise and concrete, and use a vocabulary suitable for your problem domain.
- All figures and tables in the report should be referenced and explained in the main text.
- You can use [Grammarly](#) or any LLM tool to check your grammar/spelling. Use it!
- **Make sure to remove this section and all template instructions before you submit your report!**

1 Introduction

In this part of the chapter you shall give a very brief introduction to your degree project. It shall start with a broad overview of what your project is all about. Similar to the abstract, the introduction shall make the reader interested in continue reading your report. Don't be too detailed here; there are plenty of opportunities to add details in later chapters.

Many parts of Chapter 1 can be taken from your project plan. However, give them a critical reading and try to improve them.

1.1 Background

Describe the background to the problem area of your degree project. Begin with a broad overview of what your project is all about, then focus in on the particular problem. You will need references that support your claims and the problem. Use references in IEEE format [1]. A well written background is guiding the reader from the broad perspective, why there is a need for an investigation, and narrowing down to the specific problem. First a subject area and then the area of investigation.

Important concepts that might not be familiar to the reader can also be introduced and explained here. This is also the place where you can give a brief presentation of the company/organization you are working with (if any) and motivate why they are interested in this project.

Notice: This section should not be extensive. A more thorough presentation of your problem domain will be presented in Chapter 2 (Theory).

1.2 Related work

Here you briefly describe what others have done in the field of study or how others have attempted to explain or solve the same or similar problem as you are investigating. Make very brief summaries of 2-3 papers that are directly related to your problem domain. It is okay to refer to online tech articles, blogs and portals, but you must also refer to published scientific articles.

Notice: This section should not be extensive. A more thorough presentation of related work section will be presented in Chapter 2 (Theory).

1.3 Problem formulation

Briefly describe the problem you plan to investigate, its limitations, and what results you expect. The problem formulation should be clearly related to your background and you should also point out how it relates to the related works. Try to specify the knowledge gap, the gap between what is known (related work) and what you plan to do. Although presented as a text, try to be concrete. About limitations: For example, hardware and software requirements imposed by a company. About expected results: What concrete output do you expect from the project? It can be an evaluation by controlled experiments where various approaches to solve a problem is compared, or a prototype implementation fulfilling certain criteria (e.g. accuracy > 90%).

Notice: Your project might involve handling 2-3 problems. For example: 1) Find the best possible approach to handle a certain problem and 2) Create a prototype implementing the best approach.

You can re-use parts of the problem formulation from your project plan that you have agreed with your supervisor.

1.4 Motivation

Motivate why handling the problems in your problem formulation is import. And not only for a single company/organization, why is your problem interesting for science, society or industry (in general).

You can re-use parts of the motivation part from your project plan.

1.5 Milestones

The problem you shall investigate is broken down into a list of milestones. A milestones is something you shall do within the time frame of your degree project. An milestones shall be understandable, not too small or too large, and possible to define when it is completed or not. You can read about milestones (objectives) [here](#).

M1	Milestone 1 ...
M2	Milestone 2 ...

Notice: Consider the milestone as a concrete list of smaller problems to solve (or tasks to handle) in order to handle the overall problems presented in section *Problem formulation*.

Don't mention anything about method here. It will be covered in the Chapter 3. Note that the objectives you have defined in the project plan can change slightly during the course of the project. This is not a problem. It is often difficult to foresee everything that can occur when writing the project plan.

You can re-use the motivation part from your project plan (although it is likely to be changed during the project).

1.6 Scope/Limitation

You cannot solve everything. Here you describe what you do, and what you don't do, in your project. Limitations can for example be that you only compare some frameworks of all frameworks available on the market, that you only use a specific set of software and hardware since it was a requirement from the company, that you only suggest an

architecture for a specific software product and not a general architecture, or that you only include university students in a study and not a broader population sample.

1.7 Target group

Here you outline which target group that might be interested in your work. If you, for example, do a project about software architectures, a target group can be professional developers and architects that work with similar software systems as the system you investigated. If you do a project for a company, try to identify a larger target group than only the company.

1.8 Outline

Here you outline the rest of the report. It shall contain which chapters that will follow, and what each of them is about.

2 Theory

In this chapter you should demonstrate that you have a broad knowledge of area of investigation. That is, you introduce, describe and explain the theories and important concepts to the reader, and identify and describe relevant challenges in the area of investigation and describe related work to these challenges.

This section is not about solving a problem, but it may answer one or two of your milestones. For example, the result of a literature review is often presented here (after the general introduction to the research area).

The basis for the theory chapter is most often some literature survey. Think of it as a focused textbook that introduces and explains slices of theoretical areas that are directly related to your project. Cover theories that are required for both the problem and solution, and sometimes also for the evaluation. It should be like a textbook with figures and examples. This is the chapter where your reference list grows fast!

Notice: All concepts, theories, tools, ..., you plan to use in the following chapters should be presented here. Divide the material into suitable subtopics and use one subsection for each subtopic.

2.1 Topic 1

2.2 Topic 2

3 Method

In Section 1.3 you have defined problems to investigate and milestones to handle, and you need some problem-solving activity to handle the problems. This is what we mean with a method. We have a problem, and we need some proven and structured way of approaching and solving that problem. There is no single way that works for all problems. Researchers have learned through history that particular methods are effective for problems that share some characteristics (in terms of purpose, context or problem). You can, therefore, look at how others have answered similar problems as your own problem, and use a similar method.

There is a wide range of methods you can use in your project. You can read about methods [here](#).

3.1 Research Project

In your project, as defined by the problem formulation and the milestones, it is likely that you will use several different methods. In this section we want you to state which method you plan to use to handle which problem/milestone. Motivate your decision and relate it to the problem type, stakeholders, thesis project setup, etc.

3.2 Method

Present the methods you listed in the previous section and how you plan to apply them in your project. Each method you use should be described in detail. You must demonstrate that you have a complete understanding of the methods you use. Hence, for each method you are using: 1) Describe it in general, 2) Describe how you plan to apply it in your project.

3.3 Reliability and Validity

Here you discuss the reliability and validity of your project. To answer your problem you use a method, collect (and possibly analyze) data, and draw conclusions from the data.

Reliability means if others will get the same result as you if they replicate your work. Reliability problems can, for example, occur if you use the wrong method for data collection.

It is important that you only draw conclusions that are valid, i.e. that is supported by the way you have done your work and the data you have collected.

You can read about reliability [here](#) and about validity [here](#). Discuss if you have any reliability issues or validity threats in your project here.

3.4 Ethical considerations

You are required to discuss any ethical considerations (if any) in your project. If you do an experiment you will most likely not have any ethical considerations, but in a survey ethical considerations can for example be how you make sure that the privacy of the people participating in the study is not violated (by for example removing names from the gathered data).

If you do not have any ethical considerations, state it and give a motivation.

3.5 AI disclosure

Contributions by artificial intelligence (AI) tools and technologies to the thesis project must be clearly reported. This should include the name(s) of any tools used, a description of how you used the tool(s) and evaluated the validity of the tool's outputs, and a clear statement of which aspects of the report, implementation, data or experiments were affected/generated by AI tool usage.

3.6 Team Work

This section is only relevant for students that work in a team.

The final grade for the thesis project is individual. That is, your Lnu supervisor, the reader, and the examiner must be able to say that each individual student has made a substantial contribution to the team effort. In order to facilitate individual grading each part of the work should have *one* responsible student. The team member responsible for a certain section should in some way "own" this section. That means, he/she should be able to answer detailed questions regarding this section. Also, we expect a 50-50 (roughly) division of work across the different types of project tasks. For example, both students must be involved in the technical stuff (e.g., implementation) and the more theoretical parts (e.g., writing about research methods).

Notice: Of course, you can work together on many parts of the thesis project, but only one of you can be responsible for each part of the report.

In this section you should simply assign *one* responsible student for each part of the report. It doesn't have to be lengthy. For example, Jonas is responsible for Chapters 2 and 6, and sections 1.1-1.4, 2.1, 2.3, 2.5-2.7, ...

Also, describe in text how you have divided the work between the team members. Which parts that was mainly handled by Student X, and which parts you did together.

4 Implementation

It is common that you will develop something in your project. It can be a mobile app, a stand-alone application, a website, a game, etc. In this chapter you describe the software you have implemented or the system you have created.

A good idea for how to present a non-trivial piece of software is to first present a figure showing a schematic overview of the software architecture. Try to explain the overall structure. Then dig into details in the following subsections.

Feel free to include short code snippets of important code. But not too many and not too large snippets. All such snippets should be discussed and explained in the text.

5 Experimental Setup and Results

In this section you should describe your experimental setup and results. In the ideal case you have one large experiment which results are then discussed in the next chapter (Analysis).

Notice: A project sometimes involves several experiments. In such cases it is often hard to understand the purpose of one experiment before previous experiments have been analyzed and discussed. In such cases we suggest that you merge Chapters 5 and 6 into one large chapter *Experimental Setup, Results, and Analysis* where you for each experiment present setup, results, analysis. Discuss with your supervisor about which approach to take.

By experimental setup we mean a detailed description of the environment in which the experiment was carried out. For example, what computer that was used, what software that was used, if the result to be presented is an average over 10 repeated runs, etc. In short, it should in principle be possible for a reader to repeat your experiments and get the same results if they manage to recreate your environment.

The result section should only show the raw results without any analysis, and you shall not put any conclusions or opinions in the description of the results. Try to be as objective as possible. An example of results from an experiment comparing five sorting algorithms is shown in Table 5.1 below.

Run	Bubble	Quick	Selection	Insertion	Merge
1	17384	24	3258	3	30
2	17559	21	3386	3	27
3	17795	19	3344	4	28
4	17484	20	3417	3	28
5	17642	19	3358	3	30
Average	17572.8	20.6	3352.6	3.2	28.6

Table 5.1: Execution times for the five sorting algorithms on 100 000 random numbers between 0 and 10 000.

What you show heavily depends on the type of method you use and what type of data you collect. Numerical data can for example be shown in both tables and graphs. A complementary graph for the sorting algorithms example is shown in Figure 5.1.

Note that Tables and Figures shall be labeled with chapter.number, for example Table 4.1 and Figure 1.6.

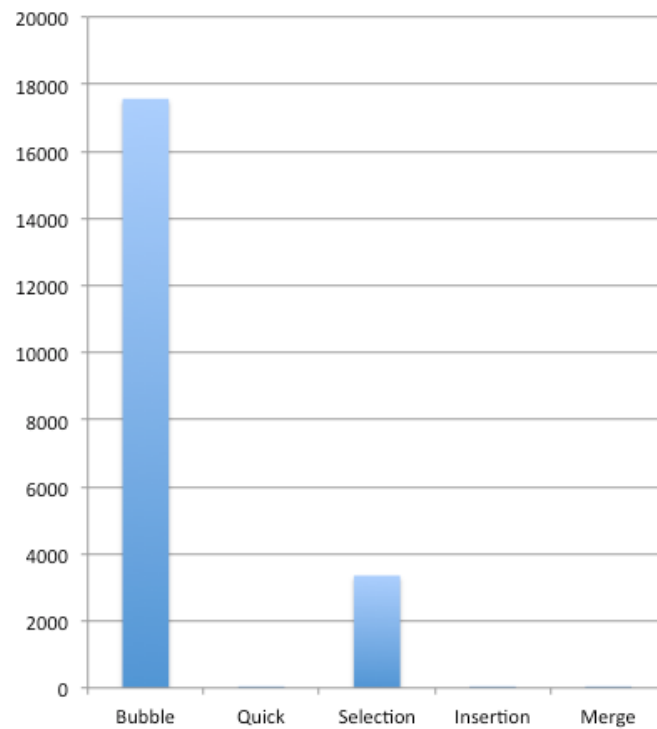


Figure 5.1: Execution times for the five sorting algorithms shown as a graph.

6 Analysis

Here you give meaning to and your own opinions of the results. What conclusions can you draw from the results? It is important that you don't draw any conclusions that cannot be backed up by your data. Consider using statistical tests to back up your claims. You can read about statistical testing [here](#).

Notice: This chapter can be dropped if you (and your supervisor) decides to take the *Experimental Setup, Results, and Analysis* approach discussed in Chapter 5.

7 Discussion

Here you discuss your findings and if your problem has been answered. Think of the project as a feedback loop. You define a problem, find a method of approaching it, conduct the study or experiment, and gather data. The data is then used to answer your problem, thus creating the loop.

You shall also discuss how your findings relate to what others have done in the field of study. Are your results similar to the findings in the related work you referenced in the report?

8 Conclusion

In this chapter you end your report with a conclusion of your findings. What have you shown in your project? Are your results relevant for science, industry or society? How general are your results (i.e. can they be applied to other areas/problems as well)? Also discuss if anything in your project could have been done differently to possibly get better results.

8.1 Future work

You cannot do everything within the limited scope of a degree project. Here you discuss what you would do if you had continued working on your project. Are there any open questions that you discovered during the project work that you didn't have time to investigate?

Here you shall include a list of all references used in your report. The reference list shall use the IEEE format. You can read about IEEE referencing [here](#). In the reference list below you can find examples of how to list a webpage [2][1], a journal article [3], a book [4] and a conference proceeding (article) [5].

References

- [1] Monash University. (2015, Oct. 13) Citing and referencing: IEEE. [Online]. Available: <http://guides.lib.monash.edu/citing-referencing/ieee>
- [2] Linnaeus University. (2015) Course Room for Degree Projects. [Online]. Available: <https://mymoodle.lnu.se/course/view.php?id=5297#section-4>
- [3] C. Lynch, “Big data: How do your data grow?” *Nature*, vol. 455, pp. 28–29, 2008.
- [4] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 3rd ed. Prentice Hall, 2010.
- [5] D. Agrawal, S. Das, and A. E. Abbadi, “Big data and cloud computing: current state and future opportunities,” in *Proceedings of the 14th International Conference on Extending Database Technology*, 2011, pp. 530–533.

A Appendix 1

In the appendix you can put details that does not fit into the main report. Examples are larger snippets of source code, long tables with raw data and questionnaires. Do not include all your source code for a project involving a lot of programming. Include only code that is important for the understanding of your project.